This is an edited version of my project writeup for an AI system I built for Outbuild with a lot more project context. You'll very likely be interested in Sections **3, 4, 6** & **7**.

1. Brief

Outbuild, a Series A construction scheduling and management SaaS company, wanted to rapidly onboard new AEs and SDRs ahead of a potential Series B. New AEs and SDRs joining the sales team faced a steep learning curve. Selling construction software to general contractors, superintendents, and project managers requires deep contextual knowledge—industry terminology, project delivery methods, lean construction principles, how pull planning works etc. — that historically took new hires 3 to 4 weeks to absorb through self learning and shadowing.

I delivered an automated learning assessment system that reduced effective onboarding time from 4 weeks to approximately 1 week by combining Claude (Anthropic's AI) for intelligent quiz generation and grading, Notion as the knowledge base and results hub, Google Forms for assessment delivery, and n8n as the automation orchestration layer connecting everything together.

The system was designed to be zero-maintenance after initial setup: any new knowledge base page added to Notion automatically gets a quiz generated, responses are auto-graded with personalized feedback, and executives receive weekly progress reports.

2. Context

2.1 Background

I came aboard to help clean up some of Outbuild's Hubspot workflows and perform data hygiene checks but saw this as a novel problem to sharpen my skills with AI tools. I had experience with Zapier but used n8n instead as the team already had prior exposure to it. Plus, getting to work with a new technology is always fun!

At the time of this project, Outbuild was a Series A company seriously considering a Series B raise, which meant aggressive hiring on the sales team. The founding team had spent a lot of time building a comprehensive internal knowledge base in Notion covering construction fundamentals, scheduling methodologies, competitive landscape, product positioning, and sales playbooks.

I use Notion as my own personal knowledge base and so felt like there was an inhouse solution that we could develop to address the learning need.

2.2 The Challenge

The existing onboarding process was essentially: read through the Notion wiki, sit in on calls, and ask questions. There was no structured assessment, no way to identify knowledge gaps, no feedback loop for the learner, and no visibility for leadership into material retention. Executive leadership also did not have the bandwidth to individually coach each hire beyond

1:1s. New hires were considered “ramped” when they felt ready, which varied wildly from person to person.

2.3 Requirements

We needed a system that could:

- **Scale to multiple users** without manual quiz creation or grading effort.
- **Cover 6 core knowledge domains in** Construction Basics, Scheduling & Planning, Project Management Solutions, Other Construction Tech, Outbuild Positioning & Messaging, and AE Playbook.
- **Provide immediate, personalized feedback** so reps know exactly what to review.
- **Give executives visibility** into team-wide onboarding progress and knowledge gaps.
- **Integrate with existing tools** (Notion, Google Workspace) rather than requiring new platform adoption.
- **Be maintainable by non-engineers** since the system should keep working as more content gets added to the KB

3. My Approach

3.1 Initial Approach vs. Final Architecture

My initial instinct was to use a web-hosted form platform like Typeform to build and manage quizzes manually. This was quickly ruled out for several reasons: manually creating 10 to 15 questions per topic across 6 domains is super tedious, questions would become stale as our KB evolved, grading and results tracking would require separate manual processes, and there was no feedback mechanism.

I went through two big iterations, refining the AI automation each time. My big win here was that the entire pipeline, from reading knowledge base content, to generating contextually relevant questions, to creating the form, to grading responses and delivering results, could be automated by using an LLM as the intelligence layer and n8n to orchestrate the flow.

3.2 System Architecture

The final system consists of three automated pipelines that operate independently:

Pipeline 1: Quiz Generation

Triggered daily (or on-demand), this pipeline scans the Notion Construction University database for any KB pages that don’t yet have quizzes. For each unquizzed page, it fetches the full content via the Notion API, extracts all text from the block structure (handling paragraphs, headings, lists, callouts, tables, and nested content), and sends the extracted text to Claude with a system prompt that instructs it to generate 10–15 multiple choice questions. Next, Claude returns structured JSON with questions, options, correct answers, explanations, and difficulty tags. A JavaScript Code node in n8n then transforms this JSON into the Google Forms API’s

batch request format (createItem objects with choiceQuestion payloads), and n8n's HTTP Request nodes execute the actual API calls to create and populate the form.

Pipeline 2: Response Grading

When a rep completes a quiz, the response arrives via a webhook. The pipeline looks up the corresponding answer key from Notion, sends both the answer key and the rep's responses to Claude for grading, writes the detailed results (score, grade, pass/fail, weak areas, recommendations) to the Notion Quiz Results database, and sends the rep an email with their results.

Pipeline 3: Executive Reporting

Every Monday, this pipeline queries all quiz results from the past week, sends them to Claude to generate an executive summary with completion rates, pass rates, average scores, individual progress tables, identified knowledge gaps, and recommendations for the onboarding team, and emails the report to the executive distribution list.

4. The Implementation

4.1 The Stack

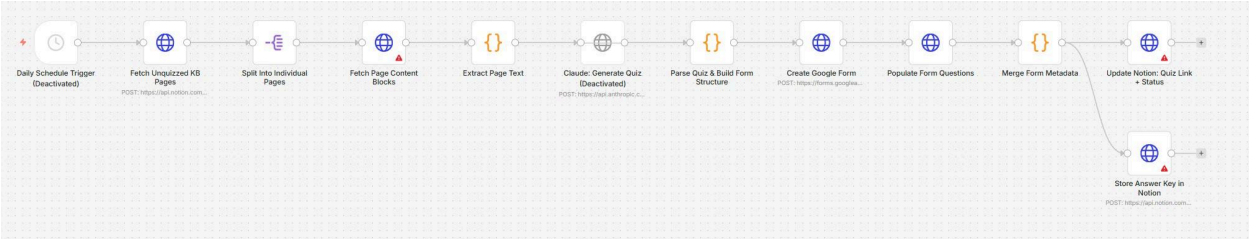
Component	Technology	Role
LLM	Claude	Quiz generation, intelligent grading, executive reporting
Orchestration	n8n	Workflow automation, API integration, scheduling
Knowledge Base	Notion API	Source content, answer key storage, results database
Assessment Delivery	Google Forms API	Quiz form creation + response collection
Response Collection	Google Sheets + Webhooks	Form response aggregation + triggering
Notifications	Email	Rep feedback + executive reports

4.2 Claude Details

I created three separate system prompts for my unique use cases in Claude.

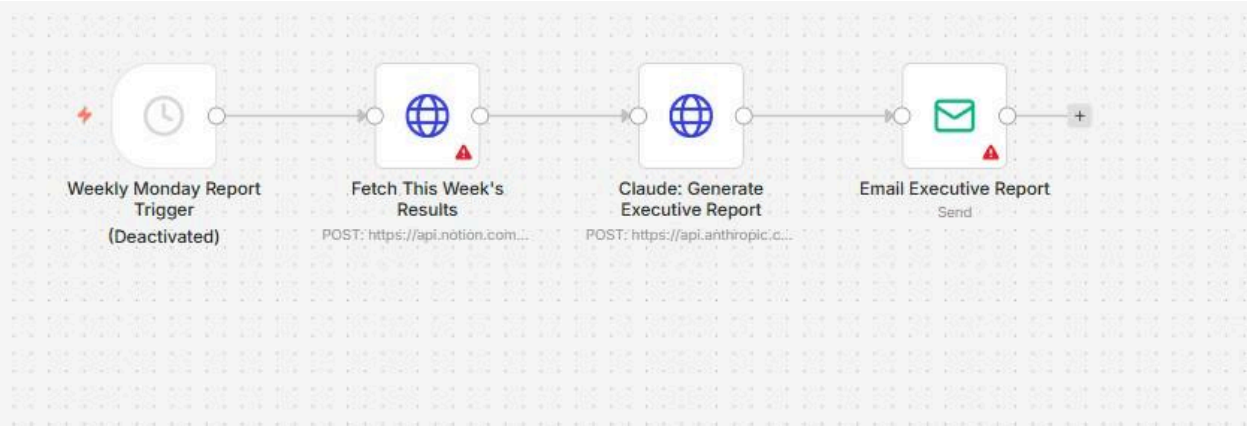
Quiz Generation: The system prompt instructs Claude to act as an onboarding quiz creator for a construction SaaS company. It specifies question count (10–15), format requirements (4 options per question), difficulty distribution (mix of foundational recall, applied knowledge, and scenario-based questions), and requires 2–3 questions per topic that specifically test how Outbuild fits into the subject matter.Claude returns pure JSON since it has

no direct access to the Google Forms API.



Grading: The prompt assigns 1 point per correct response on multi-select questions and -0.5 per incorrect response. Claude also identifies weak areas and generates learning recommendations.

Executive Reporting: Generates an HTML email report from raw results data.



4.3 Notion Schema

The system uses three interconnected Notion databases:

Database	Properties	Purpose
Construction University KB	Name, Quiz Link, Quiz Status, Questions, Total Points	Source for all KB content + tracks which pages have quizzes

Answer Keys	Topic, Form ID, Total Points, Answer Key JSON, Generated At	Stores grading rubrics
Quiz Results	Name (email), When, What (topic), Result, Grade, Passed, Recommendations	Central results

4.4 Considerations

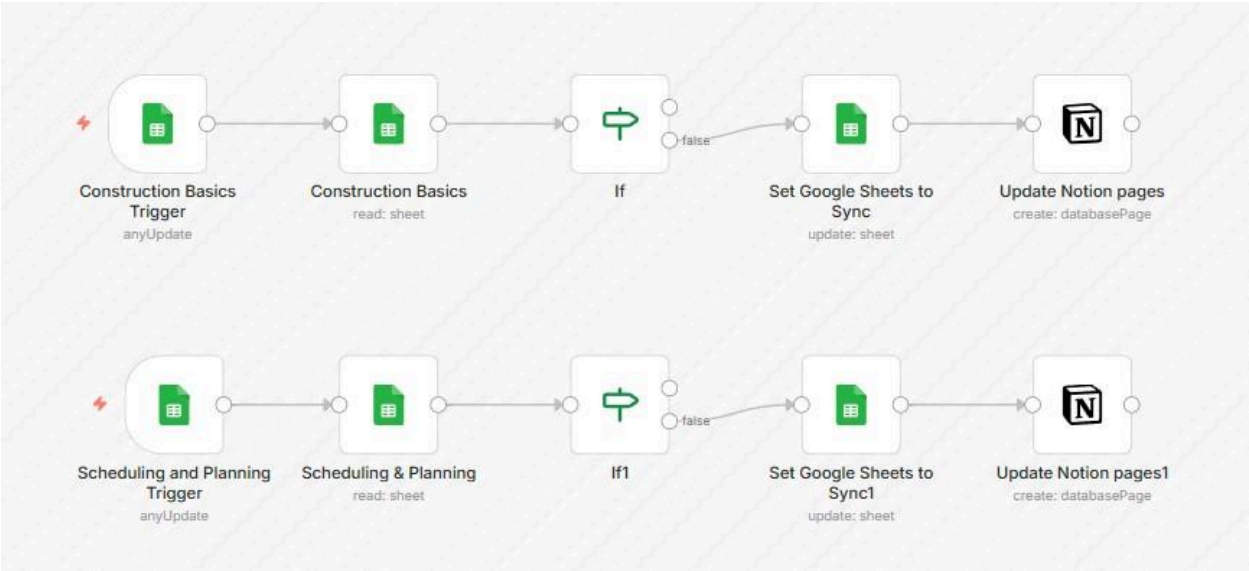
Google Forms: There were some market solutions for quiz building but I chose Google Forms because it integrates natively with Google Sheets (which the team already used), was easier for form creation, and doesn't require additional accounts for a team already on Google Workspace.

Answer keys stored in Notion, not hardcoded: My first attempt(V1) had hardcoded score denominators per topic (66, 46, 21, etc.), which broke whenever questions changed. The second pass (V2) stores the full answer key as JSON in Notion, making the grading pipeline topic-agnostic since it works for any quiz without code changes.

Unified pipeline instead of per-topic branches: The V1 workflow had 6 parallel branches (one per quiz topic), each with its own trigger, filter, sync logic, and Notion update node totaling ~36 nodes doing essentially the same thing. V2 collapses this into a single dynamic pipeline that handles any number of topics through one code path.

Webhook + polling dual ingestion: Google Forms doesn't natively push responses to external webhooks, so I used an Apps Script trigger as the primary mechanism.

5. V1 vs. V2 Comparison



The V1 prototype proved the concept was viable but had significant limitations that V2 addresses:

Features	V1	V2
Quiz Generation	Hardcoded KB content in Claude prompt	Dynamic: fetches live content from Notion API
Form Platform	Manual Google Forms	Google Forms API
Topics Supported	6 (each hardcoded separately)	Unlimited (auto-discovers new KB pages)
Grading	Raw score from Google Forms, manual denominators	Claude-powered grading with partial credit and explanations
Results Pipeline	6 parallel branches, 36+ nodes	Single unified pipeline, ~20 nodes total
Sync Tracking	Manual “Synced At” column in Sheets	Webhook
Feedback to Reps	Pass/fail Emails	Pass/fail emails + feedback
Executive Visibility	None (manual Notion checks)	Automated weekly HTML reports emailed to leadership
Maintenance	Code changes needed per topic	Zero-maintenance: new Notion pages auto-generate quizzes

6. Outcomes & Impact

6.1 Quantitative Results

- **Onboarding time:** Reduced from ~4 weeks of unstructured learning to ~1 week of guided assessment-driven learning.
- **Quiz coverage:** 6 core knowledge domains covered (Construction Basics, Scheduling & Planning, Project Management Solutions, Other Construction Tech, Outbuild Positioning & Messaging, AE Playbook) with 10–15 questions each.
- **Manual effort eliminated:** Zero manual quiz creation, grading, or results reporting after initial setup.
- **Scalability:** System designed for 20+ concurrent users with no performance impact

6.2 Qualitative Impact

- New reps received feedback on their results; reps who failed a quiz knew exactly which Notion pages to revisit, retook the quiz, and could demonstrate concrete progress.
- Sales leadership gained data-driven visibility into which topics were hardest for new hires (consistently: lean construction terminology and competitive differentiation).
- KB quality improved as quiz generation surfaced gaps and ambiguities in the source content. If Claude couldn’t generate good questions from a page, the page needed better content.

7. What I learned

- **Claude's output quality is heavily prompt-dependent.** The difference between "generate quiz questions" and the final system prompt (which specifies difficulty tiers, Outbuild-specific questions, and scenario-based formats) really determined whether or not the assessment was useful.
- **n8n's strength is connecting APIs.** Heavy JavaScript code nodes should be minimized. The V1 had chunky code blocks for text extraction but V2 keeps nodes clean and pushes intelligence to Claude.
- **Sync state is an anti-pattern when you can use webhooks.** The V1's "Synced At" column approach was chaotic and required a full read and then check cycle per response. Webhooks eliminate this wonkiness.
- **Hardcoded denominators stink!** Storing the answer key and total points alongside the form ID meant the grading pipeline could handle quiz updates gracefully.
- **Start with the executive report, work backwards.** Defining what leadership wanted to see (pass rates by topic, individual progress, knowledge gaps) clarified what I needed to see through the system.